

CHAPTER 8

Compensating Wage Differentials

Answers to End-of-Chapter Questions

1. (LO1) For this question, note the structure of the cost function. There is a positive relationship between wages and total costs, and a positive relationship between job safety and total costs. This is logical. In order to obtain the profits function, profits = total revenue (TR) – total cost (TC). The problem is simplified by the information that TR = 100.
 - a) If profits = 0, which is the case in the long run if the output market is perfectly competitive, then the isoprofit function is: $W + 10S = 100$, or in slope-intercept form, $W = 100 - 10S$. If profits = 50, then the isoprofit function is: $W + 10S = 50$, or in slope-intercept form, $W = 50 - 10S$. These two functions are linear, which is a special case. Place W on the vertical axis and S on the horizontal axis. They both have a slope of -10 and are parallel. Note that the lower isoprofit curve corresponds to higher profits, because at each level of job safety, the corresponding wage is lower on the lower isoprofit curve.
 - b) Profits = total revenue (TR) – total cost (TC) = $75 - W - 5S$. If profits = 0, then the isoprofit function is: $75 = W + 5S$. Again, we have a linear function, but it is less steep, as its slope is -5. In slope-intercept form, it is $W = 75 - 5S$. Note that this firm is safer, in that it can improve job safety more cheaply than the other firm can: at \$5 per unit rather than \$10 per unit. The two lines that correspond to the case of 0 profits will intersect, and the outermost segment (sometimes called the outer shell) constitutes the offer curve for the group of employers.
2. (LO1) This question does not contain sufficient information in order to give precise responses, so in most cases one can only assess whether the reservation wage would be lower or higher than the base case. Note that these wages are only reservation wages indicating the willingness of the worker to supply labour in each of the following cases. We can say nothing about the equilibrium wage that would prevail because we are given no information pertaining to labour demand.
 - a) If the employer provides you with a “free lunch” (note that there is no such thing as a free lunch), and you find the food of reasonably good quality, you would be willing to work for a slightly lower wage — lower by approximately the price of the food. This would constitute a negative compensating differential (meaning a wage penalty stemming from a job amenity). Therefore, the lunch is not free at all.
 - b) If the employer only required 7 hours of work per day, you would be willing to work for a lower wage — lower by your evaluation of one extra hour of leisure per day. If your wage of \$15 per hour was actually your reservation wage (this

does not necessarily apply), then the lowest wage rate that you would accept is 7/8 (7 hours out of 8) of \$15 per hour, or about \$13.13. This would constitute a negative compensating differential.

- c) Absent of any information on how much you value this total degree of flexibility, we can only say that you would probably be willing to work for a slightly lower wage. The more that you value the freedom to set your own hours of work, the greater the negative compensating differential that you would be willing to accept.
 - d) Assuming that you do not find this work to be very attractive, one would expect a positive compensating differential (meaning a wage premium stemming from a job disamenity). Absent of any information on how much you value not having to do this type of work relative to your original job description, we can only say that you would be willing to work for a higher wage than before.
 - e) Assuming that you do find this work attractive, one would expect a negative compensating differential. This depends on your personal taste for beer. Absent of any information on how much you value having to do this type of work relative to your original job description, we can say little. If you like beer, the compensating differential would be negative. If you do not like beer (and there are those of us who do not) the compensating differential would be positive.
 - f) Assuming that you do find this work attractive, one would expect a negative compensating differential. This depends on your personal taste for this type of work. Absent of any information on how much you value having to do this type of work relative to your original job description, we can probably surmise that most people would be willing to work for less. How much less depends on how much more stimulating you find this job compared to your original job (taking into consideration your taste for income as well).
 - g) Assuming that you find this work unattractive, one would expect a positive compensating differential. This depends on your personal tastes for this type of work. Absent of any information about how much you dislike having to do this type of work relative to your original job description, we can probably surmise that most people would be willing to do this kind of work only in exchange for a higher wage.
3. (LO1) First consider the case of a point on isoprofit curve I_c to the right of E_c . Imagine an indifference curve cutting through that point. In this case, the worker is not on his/her highest possible indifference curve given some wage-safety combination along I_c , which would imply a tangency with I_c . It is possible to move counter-clockwise along I_c , which makes the firm no better off nor any worse off, and makes the worker better off as he/she reaches a higher indifference curve U_c . Next consider the case of a point on indifference curve U_c to the right of E_c . Imagine an isoprofit curve cutting through that point. In this case, the firm is not on its lowest possible isoprofit curve given some wage-safety

combination along U_c , which would imply a tangency with U_c . It is possible to move clockwise along U_c , which makes the worker no better off nor any worse off, and makes the firm better off as it reaches a lower isoprofit curve I_c . Using arguments of this nature, it can be shown that any point other than E_c can be dominated by E_c , and both parties are willing to choose that point.

(See Figure 8.3 of the text)

4. (LO1) Yes, this situation is theoretically possible. Given these preferences on the part of workers, we know that the indifference curves in wage-safety space (as in Figure 8.3 of the text) are downward sloping and curved away from the origin. Imagine a firm whose isoprofit lines are horizontal. This implies that the only determinant of profitability is the wage, as job safety can be increased (decreased) continuously without sacrificing (gaining) any profitability. In effect, this would mean that it is not costly for the employer to provide a safe working environment. There would be no reason in this case to provide a risk premium, as employers should be able to hire all of the labour it wants to at the going wage (determined by factors other than the degree of job safety). The employer's offer curve would be horizontal. While job safety is valued by the employees, since it has no opportunity cost, it should not give rise to a compensating differential. If the isoprofit curves were nearly horizontal, indicating that job safety can be provided cheaply but not costlessly by firms, then one could imagine a very shallow offer curve. Equilibrium points on the right would reflect matches between the most risk-averse workers, while equilibrium points on the left would reflect matches between the least risk-averse workers.
5. (LO3) If the workers have good quality information regarding the true level of job safety, then the perceived indifference curve U_p will correspond pretty closely to the actual indifference curve U_a . It should not have much of an effect on the isoprofit curve, as this point relates to the information regarding job safety rather than the cost of providing it. The higher the level of monitoring and the more accurately that safety levels are reported, the more likely it is that both parties will reach the optimal level of safety E_0 . Remember that all of this analysis is premised upon a perfectly competitive labour market and a perfectly competitive output market, not to mention high quality information on job safety. (See Figure 8.5 of the text)
6. (LO4) The basic approach that is typically taken is to examine the situation in which the risk of a job-related death is reduced to zero. This scenario would represent a situation of a perfectly safe working environment. Since all workers would value this situation, and since employers cannot provide such an environment without cost, the labour market would assign a negative compensating differential (a wage penalty) to job safety. This differential represents the tradeoff — a product of worker choices and firm choices — between wages and job safety. This tradeoff is extrapolated (perhaps by multiplying the rate of trade off by the change in the risk from a positive probability of a fatal accident occurring to a zero probability) to a notional case of no risk at all. This product represents the market generated “price” of providing and working in the perfectly safe job, and can

be interpreted alternatively as a pecuniary valuation of the worker's life that is normally at risk.

7. (LO2, 3) This assertion refers to the rationality of the workers. According to this “wishful thinking” hypothesis, they are reasonably well informed, but they are unable and/or unwilling to behave rationally and to act in their own interests based on this information. A truly neo-classical economist does not view that scenario very likely, as they think that most people are able to act in their own best interests and do so. They have a fairly high degree of confidence in the workers. From the point of view of the government, both the case of imperfectly informed workers and “wishful thinking” may constitute grounds for intervention. Given the situation of “wishful thinking”, the best policy measure would be to provide as much information to the workers as possible about the risks that are inherent in the job. Most economists would feel more comfortable with that policy measure, because it allows the parties to reach an equilibrium on their own, which is most likely to generate an efficient equilibrium. They would tend to feel that, after accurate information has been provided to workers, it is up to them to find a job that is consistent with their aspirations for job safety and wages.

The analysis proceeds as in the subsection entitled “Imperfect Information” of the text. The effect of that policy would be to move the indifference curve of the workers such that the equilibrium that is achieved without regulation coincides with the optimal one. If a level of job safety is imposed on the firm, it might be possible to move the level of safety closer to the optimal level, but in an inefficient way. Recall that an equilibrium is inefficient if it is possible to make one party better off without making the other party worse off. Given the situation of imperfect information, the analysis again proceeds as in the subsection entitled “Imperfect Information”. The analysis of these two policy measures is pretty much the same whether the source of the market failure is imperfect information or “wishful thinking”. The case for intervention is probably stronger in the former case.

8. (LO4) The model of compensating wage differentials can be applied to any job characteristic. A family-friendly workplace practice, as a workplace characteristic, is of growing importance today given the growth in two-earner family. Firm can choose their production technology to offer workers a more family-friendly workplace (such as a more flexible hours/locations, reduced workload, support for child care...all at a cost to the firms), or they can economize on work-family characteristics and offer the savings to workers in the form of higher wages. At the same time, workers have preferences over combinations of wages and work-family balance. Obviously, they would like more of both, but at any level of utility, workers are willing to accept lower wages in exchange for some flexibility in their work schedules to achieve a balance between work and family. Not all workers have the same attitudes toward their family-friendly characteristics and will put different values on how much it worth. Workers will sort across firms according to their relative tastes for wages or for family. Those who are more family orientated will choose to work for those firms offering a more family-friendly environment, at the price

of lower wages, while those who are less concerned about it will work at the more demanding, but higher-paying jobs.

Empirical evidence on this hypothesis is limited. But some research (For examples, Gruber 1994 and Ruhm 1999) has found that mandated maternity leave has led to a drop in the wages of married women of childbearing age in both the U.S. and in Europe.

Answers to End-of-Chapter Problems

1. (LO1) In responding to this question, be careful to distinguish between 0 (i.e. zero) and O (i.e. office jobs).
 - a) In the F sector, the supply of labour is 0 for any wage below \$6. Neither type of worker will work for wages that low. At \$6, the supply of labour is infinitely elastic from $L = 0$ to $L = 100$, as all 100 of the J workers are willing to work. For wage = \$10, the supply of labour is infinitely elastic for L levels greater or equal to 100 (up to $L = 200$), as both types of workers are willing to work at that wage. For wages between \$6 and \$10, the quantity supplied of labour is still 100, so the supply is perfectly inelastic. The graph is a step function, with a vertical segment for wages between \$0 and \$6, a horizontal segment for a wage of \$6, a vertical segment for wages between \$6 and \$10 at 100, and a horizontal segment at wage = \$10 until $L = 200$. In the O sector, the supply of labour is 0 for any wage below \$6. At \$6, the supply of labour is infinitely elastic from $L = 0$ to $L = 100$. For wage = \$8, the supply of labour is infinitely elastic for L levels greater or equal to 100. For wages between \$6 and \$8, the quantity supplied of labour is 100. The graph is a step function, with a vertical segment for wages between \$0 and \$6 at $L = 0$, a horizontal segment for a wage of \$6, a vertical segment for wages between \$6 and \$8 at $L = 100$, and a horizontal segment at wage = \$8 until $L = 200$. Note that for this problem, the labour supply curve to both sectors is the same, although that is not usually the case.
 - b)
 - (i) Given these demand conditions, the equilibrium wage in the F sector is \$10, while the equilibrium wage in the O sector is \$6. The wage differential is thus \$4. As given in the problem, there are 150 workers in sector F and 50 workers in sector O.
 - (ii) 100 J workers in sector F are earning rents, because they are willing to work there for \$6, but they receive \$10.
 - (iii) There are 50 L workers working in sector F that earn no rent, because they require \$10 to work there, and that is what they are paid. All of the 50 L workers working in sector O receive no rent, as they are willing to work there for \$6, and that is what they are paid. They also earn more than their

outside opportunity in sector O, whose working conditions they like less anyway. The question is designed to show that L-type workers will tend to be employed in the O sector, and J-type workers will tend to be employed in the F sector.

2. (LO1, 4)

- a) Given this information, one would assume that there is no differential in earnings.
- b) One would expect a large increase in the demand for labour in sector A. Since the supply of labour in the short run is very inelastic because trained workers cannot be recruited quickly, one would expect a large increase in wages in sector A. This would induce more workers in sector B to quit their jobs and go through the training at their expense in order to receive the higher wages that are available in sector A. The supply of labour to sector A would increase (the long-run supply of labour is much more elastic), and the wage differential would diminish. In the long run, as the supply of labour to sector B falls, one could expect wages to increase in sector B as well.
- c) Yes. Because the employee finances the training out of her own pocket, she will expect to recover this increase in the relative cost of training for the two sectors in the form of higher wages. In effect, the supply of labour to sector B will fall relative to the supply of labour to sector A. Wages paid in sector B will rise relative to those paid in sector A.

3. (LO1, 4)

- a) In these diagrams, wage goes on the vertical axis, and paid vacations goes on the horizontal axis. Since they are both goods from the point of view of the worker, the indifference curves have a negative slope. This is because utility is increasing in both variables. Since they are both “bads” from the point of view of the firm, the isoprofit curves also have a negative slope. This is because profits are decreasing in both variables. Although both curves have negative slopes, since the indifference curves have a convex curvature and the isoprofit curves have a concave curvature, there will be an equilibrium at the point of tangency. Going from the case of one firm and one worker to the entire market, we obtain a locus of combinations of wages and paid vacations that has a negative slope. Normally it would not be a straight line, but it will have a negative slope, which indicates a negative tradeoff between the wage level and the length of the vacations.
- b) No, it would not necessarily invalidate that theory. Holding all other factors constant, one would expect a negative tradeoff between the wage level and the length of the vacations. Many other factors, such as the structure of the labour market or the structure of the product market, can influence both variables. For example, consider the case of a labour union (which implies imperfect competition in the labour market) that supplies labour to a firm that is a monopolist in the labour market. Often, the union can exercise its power in the

labour market to raise the wage above the competitive level and simultaneously improve the vacation package. These gains are financed by lower profits earned by the employer, who (in the case of monopoly) does not operate under the zero profit constraint. If two firms are totally identical in all respects pertaining to the labour market, one would expect the firm that pays higher wages to offer shorter vacations and vice versa.

- c) The wage structure would probably be compressed, which means that the differentials would diminish. Those workers who were willing to trade off some of their vacation time in exchange for higher wages would no longer be able to do so, and they would earn less than before. Those workers who were willing to trade off some of their wages in exchange for longer vacations would earn more than before. This outcome would be inefficient because some workers and firms are prevented from making mutually beneficial trades. In the absence of the regulation, it would be possible for the leisure-loving workers to be better off (and hence reach a higher indifference curve) without making their employer worse off, and for wage-loving workers to be better off without making their employer worse off.
4. (LO1, 2) The economist might argue that, provided the information on the exact level of job safety is made available to the workers and the firm (that is a big “if”), and provided that the labour market is competitive (another big “if”), the workers and the firm have reached a deal in which higher wages are paid in exchange for a lower degree of job safety than what would otherwise be the case. The firm tends to hire relatively low risk-averse workers, who are precisely those workers who value higher wages and have the highest propensity to take risks in order to receive them. By the same token, workers with such preferences are more likely to work for firms that face a relatively high cost of providing a very safe environment. The less risk-averse workers match up with the firms whose technology of production is such that they cannot provide an extremely safe environment and still be profitable. A regulation that specifies a higher degree of job safety would impose higher costs and lower profits on the firm and/or lower wages for the employees, or likely a combination of both. On the margin, the workers would rather have the higher wages than the higher level of job safety, and they are the best judge of their preferences regarding wages and job safety. Another way of saying this is that job safety is not a free good, and most of the time it comes with an expense to both parties.
5. (LO3) This quote is indeed frequently evoked by politicians from fishing areas in order to justify the expenditure levels and the generous provisions of the unemployment insurance program (at least in the case of seasonal workers) in its current state. If special EI benefits in fishing or other very seasonal industries were eliminated, it is true that many workers who currently gain their livelihood from that industry would have to exit it. It is probably an exaggeration, however, to claim that nobody could earn a living in fishing under these circumstances. The most productive and/or successful fisherman would earn the most money from fishing, and that income would help tide them over during the off-season. A

few fishers would find work during the off-season, either doing maintenance work or else in some other unrelated activity. In addition, those who cherish the lifestyle of fishing the most would be willing to accept lower wages (a negative compensating differential) than they could earn elsewhere. That point certainly applies to university professors, who generally earn considerably less income than many other occupations requiring the same (or even less) level of education. They accept lower salaries in exchange for more favourable working conditions. All in all, without the special EI benefits, we would likely see a much smaller fishery that would employ only the most attached, productive fisherman. Many but not all fishers would have to exit the industry. As far as the opinion that “a culturally important way of life would be destroyed” is concerned, most economists would consider this to be a subjective, value-laden statement. Similar claims could be made (and indeed, are made) to justify the subsidization of many activities and industries, such as dairy farming, crop farming, publishing, theatre, cinema, etc. One wonders where this list of subsidized economic activities will end.

6. (LO5)

- a) This coefficient implies that as the death rate increases (decreases) by 10,000 workers per year, the wage rate decreases (increases) by 30.03. The units for the earnings variable are not given. We are told that it refers to annual earnings, but we do not know if it refers to thousands of dollars or whatever. It is possible that the earnings are expressed in logarithmic terms, which means that 30.03 is a percentage of annual earnings. This estimate is the wrong sign, however, in light of the theory of compensating differentials, which holds that there should be a wage premium corresponding to risky jobs. This does not constitute evidence contradicting the theory, however, because the theory is stated in *ceteris paribus* terms. There are several other variables that are instrumental in determining wages that have been omitted from this regression. The effect of these omitted variables is picked up inadvertently by the estimated coefficient of the risk variable. The regression estimation process has no option but to place their influence on the back of the risk variable, which grossly distorts the true role of that risk variable. We say that that estimated coefficient is biased for the true but unknown value of beta.
- b) This is a much better specification of the earnings equation. The variables for age, education, union status, and occupation now have their own coefficients, so their estimated effects on wages, which were all being assigned to the estimated beta coefficient on the risk variable before, are now reflected in the gamma parameters, of which there are K. This coefficient implies that as the death rate increases (decreases) by 10,000 workers per year, the wage rate increases (decreases) by 43.1. The units for the earnings variable are not given. We are told that the variable refers to annual earnings, but we do not know if it refers to thousands of dollars or whatever. It is possible that the earnings are expressed as a logarithm, which means that 43.1 is a percentage of annual earnings. Note that we now obtain the expected sign, and it seems as though the original estimate was

very biased, generating an underestimate. The estimated gamma coefficients that are capturing the effects of the X variables in this part are now reflecting influences that were heaped upon the estimated coefficient of “RISK” in part a).

7. (LO5) This is a very good question. No, such a result certainly does not imply that the theory of compensating differentials is wrong in this case. There are at least two reasons why the theory has not been empirically invalidated in this case. First, there is the issue of reverse causality. As the quality-of-life index improves, according to the theory of compensating differentials, we would expect wages to fall, all other factors held constant ($\beta > 0$). On the other hand, as wages increase, that implies that many of the inhabitants are high income earners, and they are likely to purchase normal goods and services, including nice public services that are frequently associated with a high quality-of-life index. They would likely be willing to pay the higher taxes that are required to finance these services. This force would act to raise the magnitude of β , and maybe even produce a value for $\beta > 0$. This two-way causality problem (wages affecting the quality of life and vice versa) means that a simple regression estimate of β , the coefficient of the QOLI, is biased upward. Second, it is necessary to include control variables in the regression in order capture as many of the other factors that influence wages as possible, such as the level of skill, education, and training of the city’s labour force, the local cost of living, the degree of unionisation, the structure of the labour markets and output markets, etc.
8. (LO1, 4)
- a) On the income–leisure diagram, worker B has steeper indifference curves than does worker A. Since worker B values leisure relatively more, if one takes away an hour of leisure, worker B requires a greater amount of extra income in order to compensate than does worker A. Note that these indifference curves can intersect, because these are two different individuals with different preferences. The diagram is described as follows. There is one budget line that has a slope of -10 (because the wage is \$10). There is an interior solution (with a tangency) for worker B at 30 hours per work. There is an interior solution (with a tangency) for worker A at 40 hours per work, which is located 10 hours to the left of the tangency for worker B (because as we go from left to right, leisure increases and work decreases). Those two points indicate the choices that both workers would make in the absence of constraints. Since worker B is forced to work 40 rather than 30 hours and would prefer to work fewer hours, he/she is now on a lower indifference curve that passes through the budget line at 40 hours of work. Worker B’s indifference curve is steeper than worker A’s indifference curve at this point.
- b) For worker B, one would expect a large premium, as indicated by the steeper indifference curve at that point along the budget line. He/she would require a larger premium for the same amount of extra work.

- c) Within the framework for the theory of compensating differentials, firms that employ a lot of overtime hours would offer higher wages (perhaps the base wage as well as the overtime wage), *ceteris paribus*. These firms would tend to attract workers like worker A, whose indifference curves are relatively flat, with a relatively low marginal rate of substitution between income and leisure. That type of worker does not require a lot of extra income in order to be induced to put in long hours.